

# N041AV (S/N: W01041870922) — NVMe Disk Medium Error Diagnostic Report

Immfly Ground Operations / IT    Session: October 2, 2026

| Field              | Value                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Machine            | N041AV (S/N: W01041870922)                                                                                                                                       |
| Affected component | Internal NVMe disk (nvme0n1) — unrecoverable read errors                                                                                                         |
| Initial symptom    | Repeated kernel <code>critical medium error</code> messages printed on the serial console at the login prompt, roughly 111 seconds after boot.                   |
| Disk status        | <b>MEDIA ERRORS</b> — The same two sectors fail on every boot: 233560800 and 233560848 (READ).                                                                   |
| Root cause         | Persistent, reproducible read failures at fixed sectors of the NVMe disk, identical across two separate boots. Consistent with physical media degradation.       |
| Comparison         | Not a boot-sector failure (sector 0, as seen on N148AV): the unit still boots. Same pattern as N053AV, where the damaged sector also lies inside the filesystem. |
| Recommended action | <b>Disk replacement required.</b> Re-image from the fleet golden image after replacement.                                                                        |

## NARRATIVE SUMMARY

While logging in to N041AV over the serial console, the kernel printed repeated `blk_update_request: critical medium error` messages for dev `nvme0n1`, all on READ operations. The errors appeared about 111 seconds after boot, in a short burst, and always started at sector 233560800 followed by repeated retries on sector 233560848.

To check whether this was a one-off event, `grep -i error /var/log/kern.log` was reviewed. It shows the same pattern on two separate boots on October 2 (14:41 and 15:23): a burst of about ten failed reads over roughly four seconds, at exactly the same sectors, at the same point in the boot sequence. Identical sector numbers across independent boots rule out a transient glitch and point to a fixed area of the disk that can no longer be read.

The affected sectors lie well inside the disk and not at sector 0, which is why the unit still boots. Any file or data structure stored in that region will keep failing to read, and the damage may grow with use. A SMART health report was not collected during this session, so the extent of the degradation (media error count, percentage used) has not been quantified.

For completeness: the same kernel log also contains `ixgbe: probe of 0000:08:00.0 failed with error -14`, which refers to a different PCI network device and is not related to the disk. It was not investigated here.

# N041AV (S/N: W01041870922) — Troubleshooting Steps & Command Appendix

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## TROUBLESHOOTING STEPS PERFORMED

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1. **Console observation:** At the serial login prompt, the kernel repeatedly reported critical medium error on nvme0n1, sectors 233560800 and 233560848, about 111 seconds after boot.
2. **Kernel log review:** `grep -i error /var/log/kern.log` confirmed the same errors recorded in two separate boots (14:41:38 and 15:23:20), at identical sectors and at the same point after boot.
3. **Error location assessment:** Both sectors are inside the filesystem area, not sector 0, consistent with the unit still booting normally.
4. **SMART health check:** Not collected in this session. Recommended before disk replacement if the disk remains accessible.

## CONCLUSION AND RECOMMENDATION

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The NVMe disk of N041AV has persistent unreadable sectors, reproduced identically across reboots. The unit remains bootable, but the damaged region will keep producing read errors and the disk is expected to degrade further.

**Recommended action:** Issue a maintenance work order for **replacement of the NVMe disk** in unit N041AV, then re-image it from the fleet golden image. If possible, capture a SMART report before the disk is removed.

## APPENDIX: KEY COMMAND OUTPUT

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[AVA:N041AV] # `grep -i error /var/log/kern.log` (excerpt; console output is truncated at the right edge)

```
Oct 2 14:41:38 p100-ify kernel: [ 114.930866] blk_update_request: critical medium error, dev nvme0n1, sector 233560800
op 0x0:(READ) flags 0x80700 phys_seg0
Oct 2 14:41:38 p100-ify kernel: [ 114.998896] blk_update_request: critical medium error, dev nvme0n1, sector 233560848
op 0x0:(READ) flags 0x0 phys_seg 1 p0
Oct 2 14:41:38 p100-ify kernel: [ 115.095236] blk_update_request: critical medium error, dev nvme0n1, sector 233560848
op 0x0:(READ) flags 0x0 phys_seg 1 p0
... (7 more failed reads at sector 233560848, until 14:41:42)

Oct 2 15:23:08 p100-ify kernel: [ 6.392970] ixgbe: probe of 0000:08:00.0 failed with error -14

Oct 2 15:23:20 p100-ify kernel: [ 111.746821] blk_update_request: critical medium error, dev nvme0n1, sector 233560800
op 0x0:(READ) flags 0x80700 phys_seg0
Oct 2 15:23:20 p100-ify kernel: [ 111.816766] blk_update_request: critical medium error, dev nvme0n1, sector 233560848
op 0x0:(READ) flags 0x0 phys_seg 1 p0
Oct 2 15:23:20 p100-ify kernel: [ 111.913120] blk_update_request: critical medium error, dev nvme0n1, sector 233560848
op 0x0:(READ) flags 0x0 phys_seg 1 p0
... (7 more failed reads at sector 233560848, until 15:23:23)
```